Proposed First DOE for Carrot-Leaf Off-Odor Control

COEFICIENT planning paper | Version 1.0 | 17 July 2026

Purpose

This document defines a concrete, auditable first dataset for testing whether controllable LOX conditions alter early C6 aldehyde formation and whether DGMA kinetics provide useful incremental information. It also protects the protein-recovery, color, cost, and feasibility outcomes required by the COEFICIENT concept.

Planning status: proposed, not an approved laboratory SOP. Blanch duration, batch size, extraction ratio, sampling clock, analytical throughput, and final randomization must be confirmed before execution.

Design at a glance

Design element	Proposed specification
Experimental unit	One independently processed biological preparation
Blocks	Four independent carrot-leaf lots or experimental days
Conditions	Air control, pH 4, 4 deg C, N2/anoxia, blanch/LOX-off
Preparations	20 total, four per condition
Primary endpoint	Summed C6 aldehydes at T0 by validated HS-SPME-GC-MS
Critical constraints	Protein recovery, final protein content, color, chlorophyll, process burden

Scientific sequence

First quantify the untreated baseline and fraction-level protein mass balance. Then execute the mechanistic 20-preparation calibration set with fixed disruption and downstream processing. A broader optimization DOE should begin only after this campaign establishes variance, feasibility, and the validity boundary of rapid sensing.

Colored blocked design matrix

Biological block	Position 1	Position 2	Position 3	Position 4	Position 5
Lot 1	AIR DOE-001	N2 DOE-002	PH4 DOE-003	BL DOE-004	COLD DOE-005
Lot 2	PH4 DOE-006	COLD DOE-007	AIR DOE-008	N2 DOE-009	BL DOE-010
Lot 3	BL DOE-011	AIR DOE-012	N2 DOE-013	COLD DOE-014	PH4 DOE-015
Lot 4	COLD DOE-016	BL DOE-017	PH4 DOE-018	AIR DOE-019	N2 DOE-020

AIR: Air control
PH4: Acid suppression
COLD: Cold suppression
N2: N2/anoxia
BL: Blanch/LOX-off

Each colored cell is one independently processed preparation. Rows are biological lots; columns are planned positions within each block.

Detailed twenty-preparation run table

Run	Block	Order	Treatment	pH	Temp C	Oxygen	Blanch
DOE-001	Lot 1	1	Air control	Native	25	Air	No
DOE-002	Lot 1	2	N2/anoxia	Native	25	N2, sealed	No
DOE-003	Lot 1	3	Acid suppression	4.0	25	Air	No
DOE-004	Lot 1	4	Blanch/LOX-off	Native	85	Air	Yes, time TBD
DOE-005	Lot 1	5	Cold suppression	Native	4	Air	No
DOE-006	Lot 2	1	Acid suppression	4.0	25	Air	No
DOE-007	Lot 2	2	Cold suppression	Native	4	Air	No
DOE-008	Lot 2	3	Air control	Native	25	Air	No
DOE-009	Lot 2	4	N2/anoxia	Native	25	N2, sealed	No
DOE-010	Lot 2	5	Blanch/LOX-off	Native	85	Air	Yes, time TBD
DOE-011	Lot 3	1	Blanch/LOX-off	Native	85	Air	Yes, time TBD
DOE-012	Lot 3	2	Air control	Native	25	Air	No
DOE-013	Lot 3	3	N2/anoxia	Native	25	N2, sealed	No
DOE-014	Lot 3	4	Cold suppression	Native	4	Air	No
DOE-015	Lot 3	5	Acid suppression	4.0	25	Air	No
DOE-016	Lot 4	1	Cold suppression	Native	4	Air	No
DOE-017	Lot 4	2	Blanch/LOX-off	Native	85	Air	Yes, time TBD
DOE-018	Lot 4	3	Acid suppression	4.0	25	Air	No

Run	Block	Order	Treatment	pH	Temp C	Oxygen	Blanch
DOE-019	Lot 4	4	Air control	Native	25	Air	No
DOE-020	Lot 4	5	N2/anoxia	Native	25	N2, sealed	No

The displayed order is a planning sequence. Lock the final randomized order after confirming sample stability and instrument capacity.

Sampling and measurement matrix

Measurement	Starting leaf	T0 early juice	T1 post-extraction	T2 dried product
Wet mass or volume	Required	Required	Required	Required
Dry matter and total protein	Required	Required	Required	Required
DGMA time series		Required	Optional	
PTR-MS		Calibration subset	Optional	Recommended subset
GC-MS C6 compounds		Required	Optional	Recommended
L*, a*, b*	Required	Required	Required	Required
Chlorophyll	Required	Selected subset	Selected subset	Required
Sensory off-odor		Selected subset		Selected subset
Resource and cost data		Throughout process	Throughout process	Throughout process

Data architecture

Use linked tables for preparations, process events, samples and fractions, DGMA time series, GC-MS compound results, protein mass balance, color, PTR-MS, QC, and resource costs. The preparation_id is the primary join key. Raw signals and derived features must remain separate.

Decision gate

GO: mechanistic ordering and a useful T0 DGMA-to-C6 relationship. BOUNDARY: useful early prediction that does not extend to the dried product. RETHINK: weak T0 association, failed controls, or unacceptable measurement quality. Any outcome is scientifically informative.

Minimum data dictionary

Group	Field	Definition	Unit/type	Requirement
Identity	preparation_id	Stable preparation identifier	text	DOE-001
Identity	biological_lot_id	Independent leaf lot and blocking unit	text	Lot 1
Identity	sample_id	Stage-specific sample identifier	text	DOE-001-T0
Identity	parent_sample_id	Parent fraction or preparation	text	DOE-001
Identity	operator_id	Operator code	text	OP01
Identity	planned_run_order	Planned order within block	integer	1
Identity	actual_run_order	Actual execution order	integer	1
Raw material	cultivar	Carrot cultivar	text	record actual
Raw material	source	Supplier, field, or harvest location	text	record actual
Raw material	harvest_to_processing_h	Elapsed time from harvest	hours	record actual
Raw material	storage_temperature_C	Pre-processing storage temperature	deg C	4
Raw material	starting_wet_mass_g	Wet leaf mass entering process	g	record actual
Raw material	starting_dry_matter_pct	Starting dry matter	%	record actual
Raw material	starting_protein_pct_DM	Protein on dry-matter basis	% DM	record actual
Raw material	nitrogen_conversion_factor	Factor used for protein calculation	ratio	record actual
Process	target_pH	Commanded pH	pH	4.0 or native
Process	actual_pH	Measured pH after adjustment	pH	record actual
Process	target_temperature_C	Commanded treatment temperature	deg C	4, 25, or 85
Process	actual_temperature_C	Measured sample temperature	deg C	record actual
Process	oxygen_condition	Air or controlled N2 environment	category	Air or N2
Process	blanch_duration_s	Actual blanch exposure	s	TBD
Process	actual_core_temperature_C	Verified core temperature	deg C	record actual
Process	leaf_liquid_ratio	Leaf mass to liquid volume	ratio	TBD, fixed
Process	disruption_speed	Device-specific disruption setting	device unit	TBD, fixed
Process	disruption_duration_s	Disruption duration	s	TBD, fixed
Process	time_from_disruption_s	Actual sampling clock	s	record actual
Rapid sensing	dissolved_O2	Raw DGMA dissolved oxygen	instrument/raw unit	time series
Rapid sensing	dissolved_CO2	Raw DGMA dissolved carbon dioxide	instrument/raw unit	time series
Rapid sensing	K	Raw DGMA potassium response	instrument/raw unit	time series
Rapid sensing	Ca	Raw DGMA calcium response	instrument/raw unit	time series
Rapid sensing	pH_DGMA	Raw DGMA pH	pH	time series
Rapid sensing	turbidity	Raw turbidity response if available	instrument/raw unit	time series
Outcomes	total_C6_aldehydes_T0	Primary GC-MS endpoint	validated concentration basis	required
Outcomes	hexanal_T0	Individual C6 aldehyde	validated concentration basis	required

Group	Field	Definition	Unit/type	Requirement
Outcomes	OAV_index_T0	Prespecified OAV-weighted index	index	secondary
Outcomes	protein_recovery_pct	Recovered protein relative to starting protein	%	critical constraint
Outcomes	protein_content_pct_DM	Final product protein content	% DM	critical constraint
Outcomes	color_L	CIE lightness	CIE L*	required
Outcomes	color_a	CIE red-green coordinate	CIE a*	required
Outcomes	color_b	CIE yellow-blue coordinate	CIE b*	required
Outcomes	chlorophyll_total	Total chlorophyll	validated assay unit	selected subset
Outcomes	sensory_offodor_score	Calibrated off-odor score	prespecified scale	selected subset
QC and feasibility	instrument_batch	Analytical batch identifier	text	record actual
QC and feasibility	qc_flag	PASS, WARN, or FAIL	category	PASS
QC and feasibility	lod_state	Measured, below LOQ, or below LOD	category	Measured
QC and feasibility	energy_kWh	Measured process energy	kWh	record actual
QC and feasibility	water_L	Water used	L	record actual
QC and feasibility	chemical_cost_ILS	Treatment chemical cost	ILS	record actual
QC and feasibility	labor_time_min	Hands-on labor	min	record actual